Learning Route

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The solid-state drive shipped with the product contains the pre-installed system and case environment. **Before first use, confirm the mainboard bootloader version: the factory mainboard uses the JetPack 6.2 bootloader by default, while the currently shipped solid-state system is JetPack 7.2.1. You must first update the QSPI/UEFI bootloader according to the corresponding tutorial in Chapter 2, and then install the solid-state drive to boot.**

After the bootloader is updated and the system boots successfully from the shipped solid-state drive, there is generally no need to re-write the system or repeatedly build the environment already pre-installed according to the tutorial; just run the commands as required by each case. Unless there is a clear requirement, beginners should not directly perform clean system writing, system upgrade, or reinstallation steps, so as to avoid overwriting the pre-installed environment.

Chapter Introduction

It is recommended to study in the following order; the chapter numbers are subject to the current tutorial directory.

Chapter 1 Quick Start

Introduces the preparation for first use, system information, network connection, and remote access. Before the first boot, read the bootloader update instructions in "Get Started Quickly" first.

Chapter 2 Mainboard Basics

Introduces the mainboard hardware, bootloader update, system backup and recovery, system writing, and SSD expansion. Beginners usually only need to complete the applicable bootloader update and expand the SSD after entering the pre-installed system; clean system writing, image restoration, and writing to a self-provided SSD are on-demand operations.

Note:

1. Writing the Jetson clean system and component environment is an optional operation, not the first-boot process of the shipped solid-state drive. The current JetPack 7.2.1 bootloader update and clean system reinstallation are different operations; please distinguish them according to the corresponding tutorials. Reinstallation may overwrite the data and pre-installed case environment on the solid-state drive.
2. The factory image may be compressed. After successfully booting from the shipped solid-state drive, you can check and use the remaining space of the solid-state drive according to the expansion tutorial in Chapter 2.

Chapter 3 Basic Use

Mainly introduces the basic Ubuntu operations and common tools of the current system; the specific distribution is subject to the corresponding system version and tutorial description.

Chapter 4 GPIO Control

Mainly introduces the GPIO pin control, serial port, and I2C communication of Jetson Orin series mainboards.

Chapter 5 Visual Basics

Mainly introduces the simple calling of CSI cameras and USB cameras. Users can verify whether the camera is normal according to the tutorial case.

Chapter 6 OpenCV Basics

Mainly introduces common image processing in OpenCV and provides simple examples of camera calling.

Chapter 7 Advanced Visualization

Introduces the visual development environment and advanced cases such as DeepStream, TAO, YOLO, MediaPipe, and QR code recognition. Different sub-tutorials may correspond to different JetPack, CUDA, or framework versions. Check their applicable environment before starting, and do not install old dependencies directly into the current system.

Chapter 8 Docker Basics

Mainly introduces the basic usage commands and methods of Docker as well as the script startup method. The ROS1 Melodic environment runs in a Docker container; for related cases, enter the container to operate as required by the tutorial.

Chapter 9 ROS 2 Jazzy

Mainly introduces the basic knowledge and tools of ROS 2 Jazzy and robot vision cases. The ROS 2 environment runs on the host, and camera and other hardware support are subject to the description of specific cases.

Chapter 10 Large Model Basics Course

Introduces local large models, Ollama, and the basics of model deployment and application.

Chapter 11 Large Model Application Practice

Introduces the configuration and use of online and offline large models in text, speech, and multimodal applications. Online services require a network and valid credentials; offline inference speed is affected by mainboard performance.

Chapter 12 NVIDIA Isaac ROS

Introduces NVIDIA Isaac ROS features and cases. This chapter has specific JetPack, ROS 2, Docker image, and hardware requirements; check the environment baseline listed in the tutorial before starting.

Chapter 13 OpenClaw Deployment and Basic Usage

Introduces the deployment and basic usage of OpenClaw.

Chapter 14 Advanced Development of OpenClaw Extension Packages

Introduces the advanced development of OpenClaw extension packages.